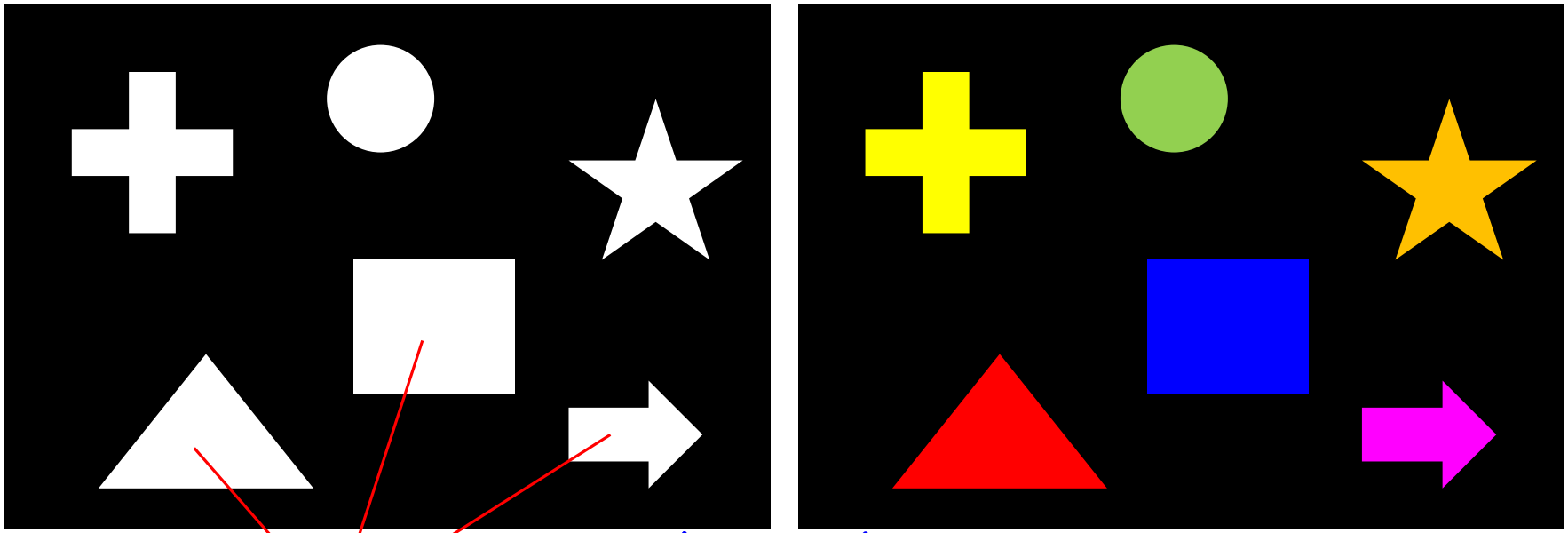
Assignment

- Connected Component Analysis (CCA)
- Due : 9. Apr (23:59)

  * No delay (score for delay is 0)

Introduction

- **What is a connected component ?**
  - **Find a connected component and assign a specific number**



"Different" connected ways

Labeling ➔ Separately handling each CC !

CCA Implementation (1/2)

- **Sequential algorithm using stack**
  - **Overall procedure is as follows :**

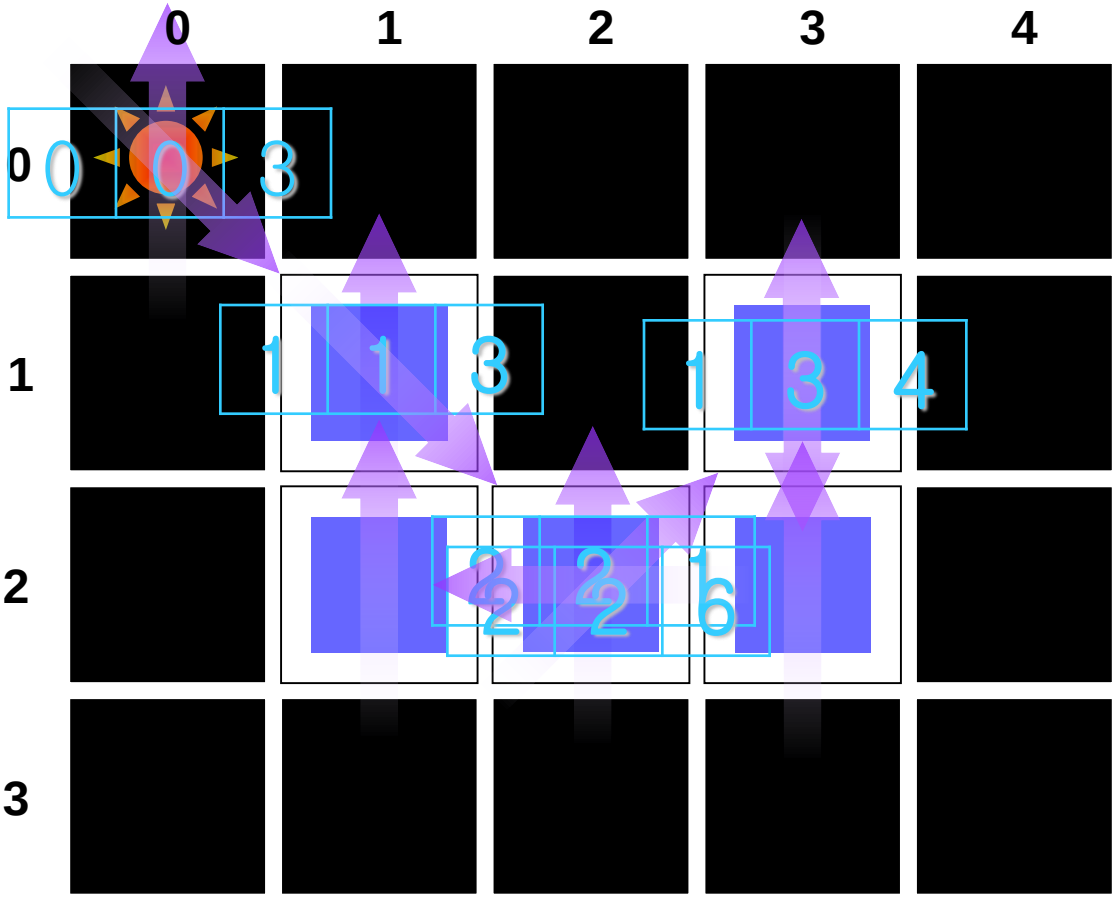
1. Scan a given 2D array from top to bottom, left to right
2. Check the label of neighbor elements at the current position
   → **Push** : when a element value of its neighbor is 1 with no label
   → **Pop** : when all the neighbor pixels already have labels
3. After scanning the whole image, generate the label map

							2		
	1	1	1				2		
	1	1	1				2		
	1	1							
	1	1			3	3	3	3	
					3			3	
					3			3	

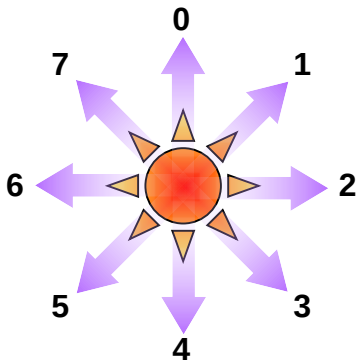
Second visit (assign 2)

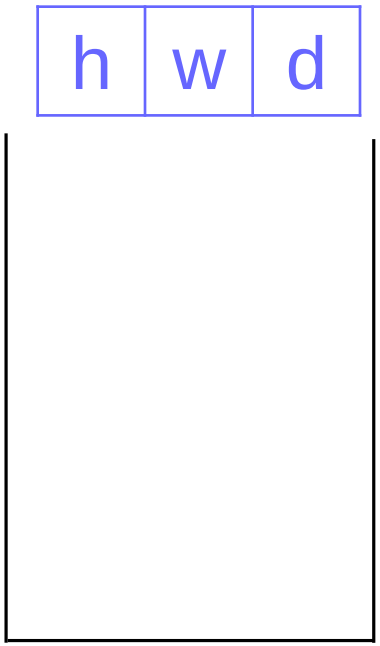
Lastly visit (assign 3)

CCA Implementation (2/2)



[Pictorial example]

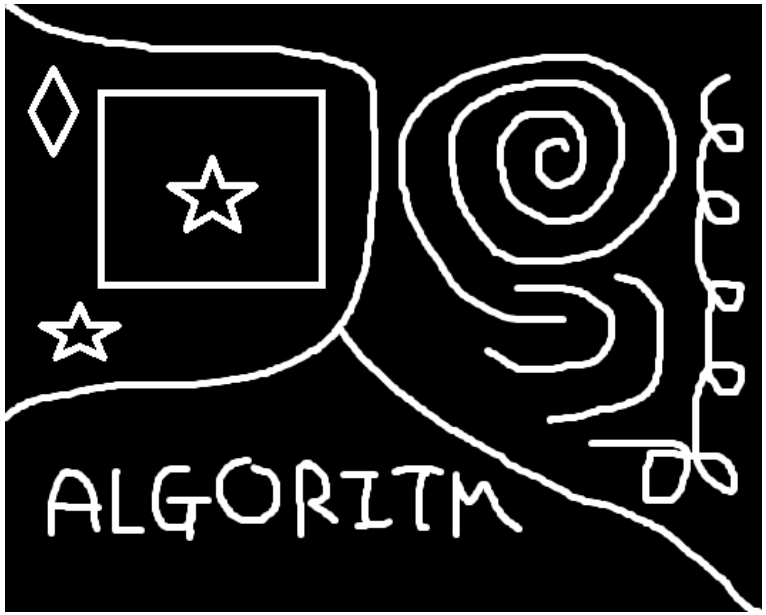




Stack

Experimental Results

- **Example for image processing**
  - **How many CC are there ?**



[Input binary image]

[Label map]

➡ **It can be used for image segmentation, object tracking, etc.**

Assignment Summary

- **Conduct CCA using the following 2D array**
  - **You must use the stack (push & pop) (do not use recursion)**
  - **Print out your label map (starting index is 2)**

```
#include <stdio.h>
#include <stdlib.h>

int input_map[15][15] = {
```

0	0	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	1	1	1	0	0	0	0	0	1	1	0
0	1	0	0	0	0	1	0	0	0	1	1	0	1	0
0	1	0	1	1	0	1	0	1	1	1	0	0	1	0
0	1	0	0	1	0	1	0	1	0	0	0	0	1	0
0	1	1	0	1	0	1	0	1	0	0	1	0	1	0
0	0	0	1	1	0	1	0	1	0	1	1	0	1	0
0	1	0	1	1	0	1	0	1	0	1	0	0	1	0
0	1	0	0	0	0	0	0	1	0	1	0	0	1	0
0	1	0	0	0	0	0	1	1	0	1	0	0	1	0
0	1	1	1	1	1	0	1	0	0	1	1	0	1	0
0	0	0	0	0	1	0	1	0	1	0	0	0	1	0
0	1	1	1	0	1	0	1	0	0	0	0	0	1	0
0	1	0	0	0	1	0	1	1	1	1	1	1	1	0
0	1	1	1	1	1	0	0	0	0	0	0	0	0	0

```
};
```